

SAFETY DATA SHEET

Creation Date 24-November-2010

Revision Date 29-March-2024

Revision Number 4

1. Identification

Product Name 12-Molybdophosphoric acid hydrate

Cat No. : 56166

CAS-No 51429-74-4
Synonyms Molybdophosphoric acid

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company

Importer/Distributor

Fisher Scientific
112 Colonnade Road,
Ottawa, ON K2E 7L6,
Canada
Tel: 1-800-234-7437

Emergency Telephone Number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

CHEMTREC Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

2. Hazard(s) identification

Classification

WHMIS 2015 Classification

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

Oxidizing solids	Category 2
Corrosive to metals	Category 1
Skin Corrosion/Irritation	Category 1 B
Serious Eye Damage/Eye Irritation	Category 1
Specific target organ toxicity (single exposure)	Category 3
Target Organs - Respiratory system.	

Label Elements

Signal Word

Danger

Hazard Statements

May intensify fire; oxidizer

Dispose of contents/container to an approved waste disposal plant

Component	CAS-No	Weight %
Phosphomolybdic acid	51429-74-4	> 95
Molybdate(3-), tetracosamolybdoxododecaphosphate(3-)-tri-hydrogen	12026-57-2	-

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.
Ingestion	Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.
Most important symptoms/effects	Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation
Notes to Physician	Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media	CO ₂ , dry chemical, dry sand, alcohol-resistant foam.
Unsuitable Extinguishing Media	No information available
Flash Point	No information available
Method -	No information available
Autoignition Temperature	No information available
Explosion Limits	
Upper	No data available
Lower	No data available
Oxidizing Properties	Oxidizer
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available

Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

Hazardous Combustion Products

Oxides of phosphorus. Thermal decomposition can lead to release of irritating gases and vapors.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

NFPA

Health
3

Flammability
0

Instability
1

Physical hazards
OX

6. Accidental release measures

Personal Precautions	Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.
Environmental Precautions	Should not be released into the environment. Do not allow material to contaminate ground water system. See Section 12 for additional Ecological Information.
Methods for Containment and Clean Up	Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.

Environmental exposure controls

No information available.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State	Solid
Appearance	Yellow
Odor	Odorless
Odor Threshold	No information available
pH	< 1 100 g/L aq.sol
Melting Point/Range	No data available
Boiling Point/Range	No information available
Flash Point	No information available
Evaporation Rate	Not applicable
Flammability (solid,gas)	No information available
Flammability or explosive limits	
Upper	No data available
Lower	No data available
Vapor Pressure	No information available
Vapor Density	Not applicable
Specific Gravity	No information available
Solubility	Soluble in water
Partition coefficient; n-octanol/water	No data available
Autoignition Temperature	No information available
Decomposition Temperature	No information available
Viscosity	Not applicable
Molecular Formula	H ₃ Mo ₁₂ O ₄₀ P . x H ₂ O
Molecular Weight	1825.25

10. Stability and reactivity

Reactive Hazard	Yes
Stability	Oxidizer: Contact with combustible/organic material may cause fire.
Conditions to Avoid	Avoid dust formation. Incompatible products. Excess heat. Combustible material.
Incompatible Materials	Bases, Organic materials, Strong reducing agents, Combustible material, Metals
Hazardous Decomposition Products	Oxides of phosphorus, Thermal decomposition can lead to release of irritating gases and vapors
Hazardous Polymerization	Hazardous polymerization does not occur.
Hazardous Reactions	None under normal processing.

11. Toxicological information**Acute Toxicity**

Product Information	No acute toxicity information is available for this product
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Component Information	No information available
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Products**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	CAS-No	IARC	NTP	ACGIH	OSHA	Mexico
Phosphomolybdic acid	51429-74-4	Not listed	Not listed	A3	Not listed	Not listed
Molybdate(3-), tetracosa-.mu.-oxododecaoxo[.mu.12-[phosphato(3-)-O:O:O:O:O:O:O:O:O:O]]dodeca-, trihydrogen	12026-57-2	Not listed	Not listed	A3	Not listed	Not listed

ACGIH: (American Conference of Governmental Industrial Hygienists)

Symptoms / effects,both acute and delayed	Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation
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Other Adverse Effects	The toxicological properties have not been fully investigated.
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12. Ecological information

Do not empty into drains.

Mobility	Will likely be mobile in the environment due to its water solubility.
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13. Disposal considerations

Waste Disposal Methods	Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.
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14. Transport information

UN-No	UN3084
Proper Shipping Name	Corrosive solid, oxidizing, n.o.s.
Technical Name	Phosphomolybdic acid hydrate
Hazard Class	8

Subsidiary Hazard Class	5.1
Packing Group	II
TDG	
UN-No	UN3084
Proper Shipping Name	Corrosive solid, oxidizing, n.o.s.
Hazard Class	8
Subsidiary Hazard Class	5.1
Packing Group	II
IATA	
UN-No	UN3084
Proper Shipping Name	Corrosive solid, oxidizing, n.o.s.
Hazard Class	8
Subsidiary Hazard Class	5.1
Packing Group	II
IMDG/IMO	
UN-No	UN3084
Proper Shipping Name	Corrosive solid, oxidizing, n.o.s.
Hazard Class	8
Subsidiary Hazard Class	5.1
Packing Group	II

15. Regulatory information

International Inventories

Component	CAS-No	DSL	NDSL	TSCA	TSCA Inventory notification - Active-Inactive	EINECS	ELINCS	NLP
Phosphomolybdic acid	51429-74-4	-	-	-	-	-	-	-
Molybdate(3-), tetracosamolybdate(3-), 12-[phosphato(3-)-O:O:O:O:O:O:O:O:O:O:O:O]dodeca-, trihydrogen	12026-57-2	X	-	X	ACTIVE	234-713-5	-	-

Component	CAS-No	IECSC	KECL	ENCS	ISHL	TCSI	AICS	NZIoC	PICCS
Phosphomolybdic acid	51429-74-4	X	-	X	X	X	X	X	X
Molybdate(3-), tetracosamolybdate(3-), 12-[phosphato(3-)-O:O:O:O:O:O:O:O:O:O:O:O]dodeca-, trihydrogen	12026-57-2	X	KE-34284	X	X	X	X	X	-

Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

Other International Regulations

Authorisation/Restrictions according to EU REACH

Not applicable

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential	Restriction of Hazardous Substances (RoHS)
Phosphomolybdic acid	51429-74-4	Not applicable	Not applicable	Not applicable	Not applicable
Molybdate(3-), tetracosa- μ -oxododecaoxo[.mu.12-[phosphato(3-)-O:O:O:O:O:O:O:O:O:O]]dodeca-, trihydrogen	12026-57-2	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Phosphomolybdic acid	51429-74-4	Not applicable	Not applicable	Not applicable	Not applicable
Molybdate(3-), tetracosamolybdoxido[.mu.12-[phosphato(3-)-O: O: O: O: O: O: O: O:]dodeca-, trihydrogen	12026-57-2	Not applicable	Not applicable	Not applicable	Not applicable

Prepared By	Product Safety Department Email: chem.techinfo@thermofisher.com www.thermofisher.com
Creation Date	24-November-2010
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Print Date	29-March-2024
Revision Summary	New emergency telephone response service provider.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS